KL Divergence

An Introduction to Kullback-Leibler Divergence

What is KL Divergence?

- KL Divergence (Kullback-Leibler Divergence) is a measure of how one probability distribution diverges from a second, expected probability distribution.
- It is used in statistics and machine learning to quantify the difference between two distributions P and Q.
- KL(P || Q) is not symmetric: KL(P || Q) ≠ KL(Q || P).

KL Divergence Formula

- For discrete distributions:

  $$KL(P \,||\, Q) = \Sigma\, P(x) * \log(P(x)/Q(x))$$

- For continuous distributions:

  $$KL(P \,||\, Q) = \int P(x) * \log(P(x)/Q(x))\, dx$$

- The value is always ≥ 0; it is 0 only when $P = Q$.

Applications of KL Divergence

- Model evaluation in machine learning.
- Variational inference in Bayesian deep learning.
- Natural language processing for measuring similarity between text models.
- Information theory to measure information loss.

Simple Example

- Let P = [0.5, 0.5] and Q = [0.9, 0.1]

$$KL(P \,||\, Q) = 0.5 * \log(0.5/0.9) + 0.5 * \log(0.5/0.1)$$

$$\approx 0.5 * (-0.152) + 0.5 * (1.609)$$

$$\approx 0.729$$

- Interpretation: Q diverges from P significantly, especially in the second value.